SABARISH G

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PROFESSIONAL SUMMARY

I solve problems domain-first: I learn how the business works, clean the data, then build the analysis or model that drives a decision. The best answer rarely comes from a fancier algorithm; it comes from asking the right business question. Quick to learn a new domain; I use AI tools to turn ideas into working products fast. Finance roots (BBA, 2 years at Wego and Jetapult), final-year MS in Business Analytics.

SKILLS

SQL & Data Cleaning:	SQL (CTEs, window functions, joins), Google BigQuery, DuckDB; data cleaning, validation and data-quality tests, data modeling (star schema)
Python & ML:	pandas, NumPy, scikit-learn, LightGBM, XGBoost, PyTorch, SHAP; classification, forecasting, model evaluation and calibration, hypothesis and A/B testing, causal inference
Advanced Excel:	XLOOKUP, INDEX-MATCH, SUMIFS / COUNTIFS, FILTER / UNIQUE / SORT, LET, Pivot Tables, Power Query, Power Pivot, data validation, VBA macros
Dashboards & AI:	Tableau, Looker, Power BI, Streamlit; multi-agent LLM systems (LangGraph, MCP), AI-assisted coding, rapid prototyping; dbt, Google Cloud

PROJECTS

Flight Delay Prediction on 39M Flights: Domain Data Beats Model Complexity

Python | LightGBM | SHAP

Code (GitHub): github.com/sabarishknight/regime-aware-flight-delay-prediction

- Audited 65 raw monthly files (date corruption in 36, schema drift in 7) and validated one time zone clock across 387 airports.
- Predicted delays and cancellations 2 hours ahead (PR-AUC 0.748 vs 0.350 baseline, calibrated on an unseen year); incoming-aircraft data added 21.7% accuracy and a costlier model 0.3%, so I shipped the simpler model and cut expected alerting cost up to 55%.

Credit Policy Optimizer: Approve, Refer or Decline on 789K Real Loans

Python | scikit-learn | Streamlit

Code (GitHub): github.com/sabarishknight/credit-policy-optimizer

- Built a credit scorecard on 789K real loans and tested it on 81K unseen loans (AUC 0.730, Gini 0.46); it beat a gradient-boosting challenger and the lender's own grade (0.689), so I kept the simpler, explainable model.
- Turned scores into an approve / refer / decline policy: profit up from €2.0M to €4.7M, defaults on approved loans down from 8.1% to 4.7%, 70% still approved, and a reason code for every decline.

Metric On-Call: Multi-Agent AI That Explains Why a KPI Moved (in progress)

LangGraph | MCP | BigQuery | dbt

Code (GitHub): github.com/sabarishknight/metric-on-call

- Designing 8 AI agents that run a senior analyst's root-cause investigation, from data audit and KPI breakdown to dollar impact; tested SQL and Python do the maths, and a Skeptic agent challenges every answer.
- Testing on 13 planted incidents with known causes (pipeline bugs, cannibalisation, Simpson's paradox) plus quiet weeks.

FaceRoll: Face-Recognition Attendance App, Used Daily by 2 Professors

Python | Flask | Computer vision | Cloud Run

Code (GitHub): github.com/sabarishknight/faceroll-attendance

- Built and deployed an app my MSc class uses daily: one class photo marks attendance, with guided phone enrolment, review of unclear matches and Excel export.

RESEARCH

CF-NODE: Counterfactual Spatial Node Intervention for Test-Time Adaptation Under Domain Shift

Paper (arXiv): arxiv.org/abs/2610.22437 | **Code (GitHub):** github.com/sabarishknight/CF-Node

- Designed a training-free method that keeps diabetic-retinopathy grading models reliable on images from new hospitals and cameras, with no retraining; best-in-class AUC and Macro-F1 against four state-of-the-art methods on 5 external datasets.

PROFESSIONAL EXPERIENCE

Finance Associate · Wego

Jun 2023 – Nov 2024

Online travel search platform · Bengaluru · Converted to full-time from Finance Intern (Jan – May 2023)

- Cleaned, mapped and validated two years of financial records (BigQuery, SQL, Excel) for the SAP to Oracle NetSuite migration, with the heads of Product, Engineering, Marketing and HR.
- Built Tableau and Looker dashboards on booking volume, cost per booking and profitability by product and region for leadership.
- Automated recurring reports with VBA and BigQuery and redesigned the workflow with Six Sigma DMAIC, cutting 27% of process steps and saving 30+ hours a week.

Finance Associate · Jetapult

Nov 2024 – Feb 2025

Mobile gaming venture studio · Bengaluru

- Built leadership reports combining user-acquisition and game data, and prepared the data behind the investor pitch deck.

EDUCATION

MS, Business Analytics (final year) · Manipal Academy of Higher Education

Jul 2025 – May 2027

Machine Learning, Predictive Analytics, Statistical Modeling, Data Visualization, Big Data Analytics

BBA, Finance · PES University, Bengaluru

Aug 2020 – Aug 2023

CERTIFICATIONS

IBM Data Science (2026) · Microsoft AI Product Manager (2026) · Anthropic: Model Context Protocol, AI Fluency · Google Project Mgmt